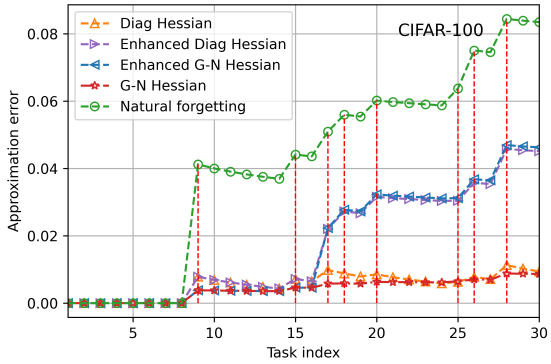
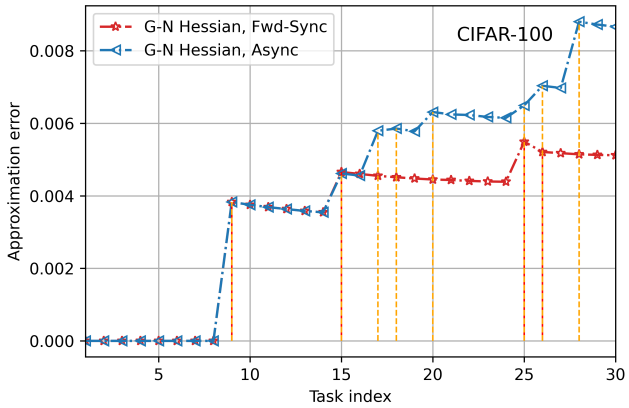
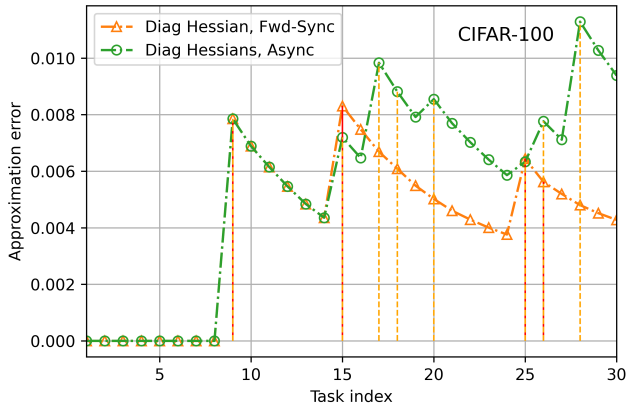


Figure A: Evolution of approximation error $\|w_t^{-\mathbf{S}_{1:t}} - w_t^{-S_{\leq t}}\|$ across 30 tasks in CLU process on CIFAR-100 dataset, under the following unlearning sequence: $(9, \{3, 4, 6, 8\}), (15, \{11\}), (17, \{12\}), (18, \{13\}), (20, \{14\}), (25, \{23\}), (26, \{20, 25\}), (28, \{24, 10\})$.

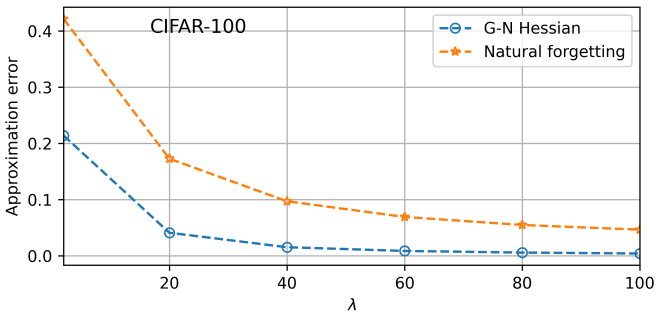


(a) Approximation error $\|w_t^{-\mathbf{S}_{1:t}} - w_t^{-S_{\leq t}}\|$ over task index t using Alg. 2 with the Gauss-Newton Hessian.

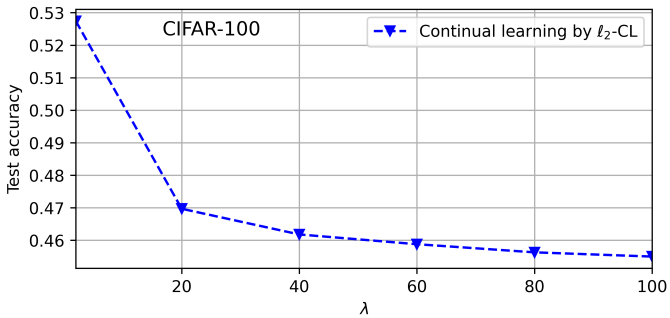


(b) Approximation error $\|w_t^{-\mathbf{S}_{1:t}} - w_t^{-S_{\leq t}}\|$ over task index t using Alg. 2 with the diagonal Hessian.

Figure B: Evolution of approximation error $\|w_t^{-\mathbf{S}_{1:t}} - w_t^{-S_{\leq t}}\|$ over t, under (a) Alg. 2 with Gauss-Newton Hessian and (b) Alg. 2 with the diagonal Hessian, under forward-synchronous and asynchronous unlearning sequences. In (b), the approximation error under the diagonal Hessian is higher than or equal to that under the asynchronous sequence at times 15 and 25, because more tasks have been unlearned by then in the forward-synchronous sequence. However, once both sequences have unlearned the same set of tasks, the forward-synchronous sequence yields a lower approximation error than the asynchronous sequence.



(a) Final approximation error after unlearning



(b) Final test accuracy after CL without unlearning

Figure C: (a) Final approximation error $\|w_T^{-\mathbf{S}_{1:T}} - w_T^{-S_{\leq T}}\|$ versus the regularization parameter λ, after unlearning by Alg. 1 and Alg. 2. (b) Final test accuracy versus the regularization parameter λ, after training by the ℓ_2-CL algorithm without unlearning.